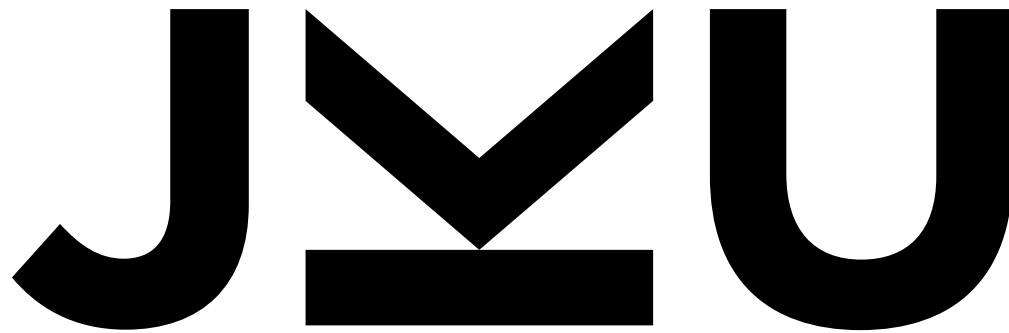
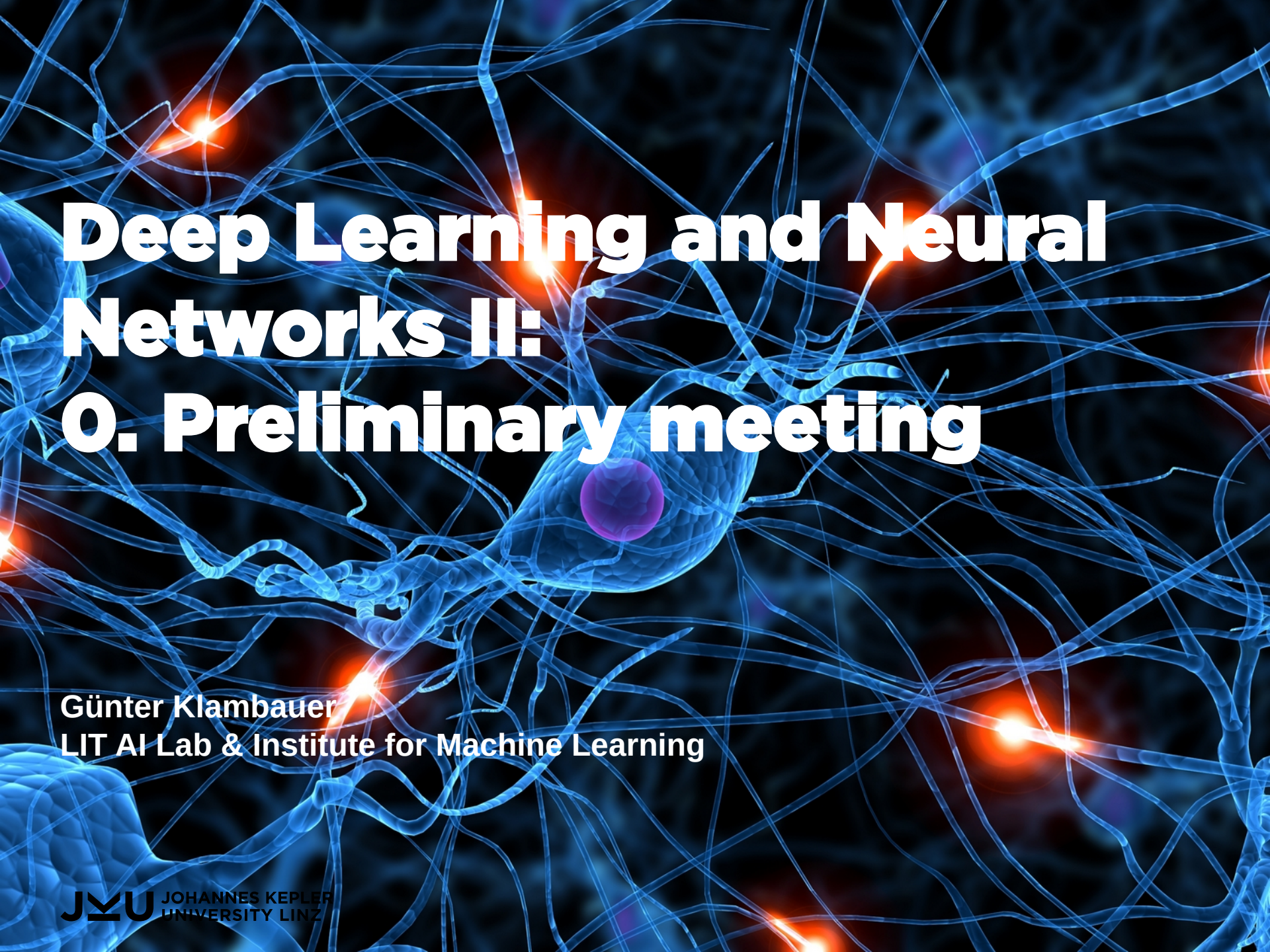




**JOHANNES KEPLER
UNIVERSITY LINZ**



Deep Learning and Neural Networks II:

0. Preliminary meeting

Günter Klambauer
LIT AI Lab & Institute for Machine Learning

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Modus operandi – Course overview

- Basic course of Master Artificial Intelligence
- Class:
3 ECTS 2 SWS VO Fr 10:15 – 11:45
- Exercise: 1.5 ECTS 1 SWS UE
- **Materials:**
 - _ Lecture notes will be online on Moodle (finished early April)
 - _ Slides will be made available via Moodle
 - _ Video recordings available on Moodle for at least two weeks
- **Exams:**
 - _ Class: Written exam on 27th June 2025 (written exam)
 - Re-examination: Sept 2023 (TBA)
 - _ Exercise: Bi-weekly assignments (evaluated)
- **Literature:**
 - _ Lecture notes DLNN1
 - _ Deisenroth, M. P., Faisal, A. A., & Ong, C. S. (2020). Mathematics for machine learning. Cambridge University Press.
 - _ Bengio, Y., Goodfellow, I., & Courville, A. (2017). Deep learning (Vol. 1). Massachusetts, USA:: MIT press.
 - _ Bishop, C. M. (2006). Pattern recognition and machine learning. Springer.
- **Contact:** dlenn@ml.jku.at

Content of this lecture (subject to change)

1. Recap and basics
2. Advanced CNN architectures (“ConvNets II”)
 1. Image classification and recognition
 2. Semantic segmentation
 3. Object detection
3. Generative and unsupervised DL methods
 1. Autoencoders
 2. Variational Autoencoders
 3. GANs
 4. Flow-based generative models
 5. Diffusion models
4. Bayesian Deep Learning
5. Graph neural networks
6. Deep Learning Theory

Overview (subject to change)

DATE	CONTENT
7.3.	Recap
14.3.	Image Classification
21.3.	Object detection and localization
28.3.	Semantic segmentation
4.4.	Autoencoders
2.5.	VAEs
9.5.	GANs, W-GANs
16.5.	Metrics (FID), Normalizing Flows
23.5.	Diffusion Probabilistic Models
30.5.	Bayesian Deep Learning and Uncertainty
6.6.	Graph Neural Nets
13.6.	Theory
27.6.	Exam

Notation is extended a bit

- We will also use h for the activations in a hidden layer; before: a
- We will sometimes use $g_w(x)$ instead of $g(x; w)$ if the focus is not on the weights
- We will also use $p_\theta(x)$ if a neural network that models a probability

Summary

- Lecture will cover relevant, recent and active Deep Learning areas
- Exercises will support those lectures; implementation of architectures and learning methods in Pytorch
- Final online exam at end of the lecture

Master study AI 2nd semester

Timetable: Master "Artificial Intelligence", 2nd Semester, Summer Term 2024										
	Monday		Tuesday		Wednesday		Thursday		Friday	
08:00										
09:00						08:30 VL AI and Law II HS 6 136.307	08:30 UE AI in Life Sciences HS 6 365.241			
							09:15 simult.DL and NNII 365.247			
							09:15 UE Artificial Intelligence in Life Sciences 365.257			
10:00						10:00	10:00			
	10:15 KV Introduction to autonomous vehicles HS 7 / Track 1 <i>blocked</i> 245200		10:15 VL Artificial Intelligence in Life Sciences Track 4, 365.240		10:15 KV Robopsychology HS 6 510.006		10:15 VL Theoretical Concepts of Machine Learning HS 7 365.041		10:15 VL Deep Learning and Neural Nets II HS 6 365.111	
11:00			11:00				11:00 UE Deep Learning and NN II <i>blocked</i> 365.248			
	parallel:				11:45		11:45		11:45	
12:00	12:00 KV Genome Analysis & Transcriptomics HS 6 / Track 4 365.109		12:00 VL Statistics for AI HS 7 366.590		12:00 VL Introduction to Robotic Systems HS 6 / Track 1 370.002					
13:00	13:30		13:30		13:30		12:45 UE Introduction to Robotic Systems, Track 1 370.003			
14:00	13:45 UE Deep Learning and Neural Nets II <i>blocked</i> HS 6 365.114		13:45 UE Deep Reinforcement Learning HS 6 365.250/.251 Group A & B		13:45 KV Structural Bioinformatics HS 6 / Track 4 365.110		13:45 UE Theoretical Concepts of Machine Learning 365.042/365.244 365.100/365.245 <i>blocked, all groups parallel</i>			
15:00	15:15		15:15		15:15		15:15			
16:00	15:30 VL Machine Learning: Unsuperv. Techniques HS 7 365.077		15:30 VL Knowledge Representation and Learning HS 6 351.009							
	UE Deep Learning and NN II <i>blocked</i> 365.246									
17:00	17:00		17:00							
18:00			17:15 UE Knowledge Represent- ation and Learning HS 6 351.010				17:15 VL Deep Reinforcement Learning HS 6 365.249			
							18:45			